

Bikal Shrestha

Spatial Data Analyst & GIS Researcher

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SUMMARY

Spatial data analyst and GIS researcher with dual master's degrees in Geographic Information Science and Data Analytics from Clark University. Research sits at the intersection of deep learning, machine learning, remote sensing, and environmental science — building end-to-end pipelines from satellite imagery to production-ready interactive dashboards.

EDUCATION

MS in Geographic Information Science Clark University, Worcester, MA	Aug 2024 – May 2026
MS in Data Analytics Clark University, Worcester, MA	2021 – 2023
BSc in Computing Coventry University	2016 – 2019

EXPERIENCE

Data Analyst / Backend Developer Softwarica College of IT and E-commerce	July 2020 – July 2021
- Cleaned millions of records across 250 Moodle database tables using SQL and Tableau Prep - Built executive, marketing, and performance dashboards in Tableau for institutional reporting - Designed and shipped an Android app and REST APIs using Flutter and Node.js	
Teaching Assistant Softwarica College of IT and E-commerce	June 2019 – July 2020
- Supervised ~40 undergraduate students per semester on independent research projects - Mentored teams building hardware/software projects including an electric vehicle prototype	
Staff Manager Hotel Pokhara Peace	Jan 2017 – Sept 2019
- Managed daily operations and staff coordination across departments - Developed an internal web presence using HTML, CSS, and JavaScript	

PROJECTS

Dual-Branch U-Net for Precipitation Downscaling PyTorch · Python · NASA IMERG · Google Earth Engine · DEM Downscales NASA IMERG from 10 km to 250 m over Hawaii using dual-branch CNN fusing satellite and topographic data. Test loss 0.0332 with 2.3M parameters.
Nepal GLOF Explorer Streamlit · scikit-learn · Google Earth Engine · Sentinel-2 · Landsat Tracks 25 glacial lakes across Nepal Himalaya (2000–2024) with ML-based GLOF risk classification and satellite change detection.
California Wildfire Analysis Dashboard

Streamlit · Plotly · Pandas · MTBS/USGS · Docker

38-year (1984–2022) interactive dashboard of California wildfire history from MTBS satellite imagery and climate records.

MappingAfrica — Satellite Agricultural Field Segmentation

PyTorch · UNet · FastAPI · React · rasterio

Semantic segmentation of farmland in Zambia using multi-spectral imagery. Achieves 81.79% pixel accuracy and 43.31% mIoU.

Peru Wildfire Dashboard

Next.js · MapLibre GL · Python · GeoPandas · NASA FIRMS · MODIS

Visualizes 24 years of fire hotspot and burned area data across Peru with 32,000+ NASA FIRMS hotspots and land governance analysis.

SKILLS

Languages	Python, R, SQL, JavaScript, Node.js	Deep Learning	PyTorch, UNet, CNN, Transfer Learning, Semantic Segmentation
Machine Learning	scikit-learn, Random Forest, DBSCAN, Clustering, Regression	Data & Visualization	Pandas, NumPy, Streamlit, Plotly, Tableau, Chart.js
GIS & Remote Sensing	Google Earth Engine, QGIS, ArcGIS, TerrSet, Sentinel-2, Landsat, NASA IMERG, GDAL	Research Areas	Climate Modeling, Hazard Mapping, Wildfire Analysis, Precipitation Downscaling